HEALTHSPHERE AI CLINICAL DECISION SUPPORT

CRITICAL PATIENT EMERGENCY REPORT & RISK ASSESSMENT

Patient Name: Robert Vance | **MRN:** MRN-100012 | **Age/Gender:** 64 / Male

Blood Group: O+ | **Primary Condition:** Hypertensive Crisis & AKI | **Status:** RED FLAG EMERGENCY

1. ACUTE CRITICAL VITALS & BIOMARKERS

Biomarker / Vital Parameter	Recorded Value	Reference Range	Clinical Status
Blood Pressure (Systolic / Diastolic)	195.0 / 115.0 mmHg	90-120 / 60-80 mmHg	**CRITICAL HIGH (Stage 3 Crisis)**
Heart Rate	110.0 bpm	60-100 bpm	**Tachycardia (High)**
Fasting Blood Glucose	248.0 mg/dL	70-99 mg/dL	**Severe Hyperglycemia**
Serum Creatinine	3.4 mg/dL	0.7-1.3 mg/dL	**Acute Kidney Injury (3x Normal)**
Total Cholesterol	290.0 mg/dL	< 200 mg/dL	**Severe Hyperlipidemia**
Body Mass Index (BMI)	30.5 kg/m²	18.5-24.9 kg/m²	**Class I Obesity**

2. AI AGENT RISK ENGINE & SHAP EXPLANATION

Target Condition	Predicted Probability	Severity Tier	Key SHAP Risk Drivers
Cardiovascular / Heart Disease	80.0%	CRITICAL RISK	Systolic BP (195), Age (64), Cholesterol (290)
Type 2 Diabetes Mellitus	65.0%	HIGH RISK	Glucose (248), BMI (30.5)
Chronic Kidney Disease (CKD)	65.0%	CRITICAL RISK	Creatinine (3.4), Systolic BP (195)

3. HUMAN-IN-THE-LOOP RED-FLAG ESCALATION & AGENT ACTION

Patient Reported Message: "This sounds like it could be urgent. I'm not able to assess symptoms like this myself, so I've flagged it for a clinician to reach out right away. If this is a medical emergency, please contact emergency services or go to the nearest emergency room immediately."

AI Safety Gate Triggered: Red-Flag Symptom Gate (Emergency Urgency)
Escalation Ticket Created: Ticket #1 (Category: Clinical / Physician Worklist)
Agent Handoff Action: Immediate alert sent to attending physician worklist; triage protocol initiated.

4. CLINICIAN ASSIST AGENT BRIEF & DRAFT CARE PLAN

AI Chart Prep Brief: Patient: Robert Vance (64y, Male, primary condition: Hypertensive Emergency, Type 2 Diabetes, Acute Renal Impairment) Latest risk assessment (2026-10-06): overall High, heart 80.0%, diabetes 65.0%, CKD 65.0%. ■ 1 open escalation ticket(s): Urgent symptom description: "I have severe crushing chest pain, shortness of bre

Immediate Emergency Directives:

1. Immediate ICU Triage & Continuous Cardiac Monitoring.
2. Administer IV Labetalol / Nicardipine for controlled BP reduction.
3. Start IV Insulin protocol for glycemia control (Glucose 248 mg/dL).
4. Stat Nephrology consult for Acute Kidney Injury (Creatinine 3.4 mg/dL).
5. Avoid nephrotoxic NSAIDs or contrast media.

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